

Lance DeCandia

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EDUCATION

Stevens Institute of Technology - B.S. Computer Science - Expected May 2027

Hoboken, New Jersey

Relevant Coursework: Machine Learning, Data Mining & Knowledge Discovery, Data Structures, Algorithms, Databases, Computer Architecture, Embedded Systems, Operating Systems, Cybersecurity, Networking (TCP/IP), DevSecOps, Object-Oriented Programming, Software Development

AP Coursework: AP Environmental Science (high school), AP Computer Science A (high school)

AWARDS & HONORS

Edwin A. Stevens Scholarship - Stevens Institute of Technology

Dean's List - Stevens Institute of Technology

EXPERIENCE

Flox - DevOps Engineer, New York, NY

May 2026 - Present

- Ported a full-stack capstone project from Docker Compose to a fully automated Flox/Nix developer environment, resolving Python venv/PATH ordering, Redis version pinning, and cross-platform port conflicts across macOS, Debian, and Ubuntu
- Benchmarked CI/CD build performance across GitHub Actions and Blacksmith runners across multiple caching strategies and workload sizes, then authored an engineering blog post and documentation recipes based on the findings to drive an internal CI/CD upgrade
- Evaluated 17+ sandbox platforms against a 7-point rubric assessing Nix/Flox integration readiness, producing a ranked shortlist and research documentation
- Built proof-of-concept Flox environments for real-world projects and published them to FloxHub with walkthrough documentation, serving as demos to present Flox's capabilities to other developers and teams

PROJECTS & RESEARCH

WatchWeave - Anime Tracking Platform (DevSecOps)

March 2026

- Built a full-stack anime tracking application in a 3-person team using Node.js, Express, React, and PostgreSQL, deployed on AWS Elastic Beanstalk with Docker containerization
- Designed and compared Redis (AWS ElastiCache) vs. in-memory (LRU) caching strategies for Jikan API integration, measuring performance via Prometheus + Grafana dashboards
- Implemented dual CI/CD pipelines on GitHub Actions and GitLab CI/CD with Trivy container scanning and Snyk dependency scanning for security automation

Flight Delay Prediction System

December 2025

- Built binary classification models on 100,000+ records from a 2019 aviation dataset including weather, aircraft, airport, and historical flight data
- Conducted extensive exploratory data analysis and feature engineering; implemented and compared KNN, ANN, Naive Bayes, Random Forest, and K-means clustering
- Evaluated model performance metrics across all algorithms to identify the most effective approach for real-world delay forecasting

FreeRTOS Embedded Systems - Real-Time Scheduling (C)

November 2025

- Implemented a countdown timer and multitasking scheduler on embedded hardware using FreeRTOS, applying real-time scheduling algorithms including EDF, RM, and DM
- Analyzed task scheduling tradeoffs and verified timing constraints in a resource-limited embedded environment

Breast Cancer Classification - Data Mining Project

October 2025

- Applied SVM, Self-Organizing Maps (SOM), C5.0 decision trees, and Random Forest to classify malignant vs. benign tumors on a real clinical dataset
- Performed data preprocessing, dimensionality reduction, and model comparison with detailed accuracy, precision, and recall analysis
- Demonstrated how machine learning can surface actionable insights from complex, noisy medical data

File System Implementation - Linux (C)

May 2025

- Collaborated with a 3-person team to design and build a fully functional file system for Debian Linux using C
- Engineered inode-based file management, block allocation, directory structures, and core file I/O operations

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, R, Bash, Assembly, OCaml, HTML/CSS, Ruby, Nix

Data & ML: scikit-learn, Pandas, NumPy, Jupyter, EDA, classification, regression, clustering, model evaluation

Tools & Platforms: Linux, Git, Excel, Logisim, FreeRTOS, Docker, Jira

LEADERSHIP & EXTRACURRICULARS

Stevens Computer Science Club

January 2024 - Present

- Collaborated with student organizations to host tech events for 100+ participants; mentored peers in Python and Java and led tutoring sessions on data structures and algorithms

Stevens Competitive Ice Hockey - Forward

August 2023 - Present

- Competed at a high level throughout college; developed strong teamwork, communication, and performance under pressure across years of team competition

Volunteer - St. Vincent De Paul Furniture Ministry

June 2019 - Present

- Delivered furniture to families and individuals in need throughout the local community, providing direct support to underserved residents